

Video Link 1: [https://youtu.be/V\\_QS1HzJ8Bc?si=dUoUcQvIM9GmrYOq](https://youtu.be/V_QS1HzJ8Bc?si=dUoUcQvIM9GmrYOq)

Video Link 2: <https://youtu.be/OxaYvJquPe0?si=Bv6p4B5uYFGcBh63>

### Cache Mapping Techniques — Short Exam Version

| Feature          | Direct Mapping           | Associative Mapping          | Set-Associative Mapping               |
|------------------|--------------------------|------------------------------|---------------------------------------|
| Address Fields   | Tag, Block, Word         | Tag, Word                    | Tag, Set, Word                        |
| Comparison       | Only <b>1 comparison</b> | Compare with <b>all tags</b> | Compare with <b>blocks in the set</b> |
| Mapping Location | One fixed cache location | Any cache block              | Any block within a specific set       |
| Search Time      | <b>Lowest / Fastest</b>  | <b>Highest / Slowest</b>     | Moderate                              |
| Tag Bits         | Least                    | Most                         | Between Direct and Associative        |
| Cost             | Low                      | High                         | Moderate                              |

## 1. Direct Mapping

### Advantages:

- Simple and easy to implement.
- Fast because only one tag is checked.
- Less expensive.

### Disadvantages:

- Lower hit ratio due to frequent replacement.
- Two blocks mapping to the same location may cause **conflict misses**.

## 2. Associative Mapping

### Advantages:

- Flexible mapping.
- High cache hit ratio.
- Easy to search using tag comparison.

### Disadvantages:

- Expensive.
- Requires comparison with all cache tags.
- More complex hardware.

## 3. Set-Associative Mapping

**Advantages:**

- Combines the benefits of direct and associative mapping.
- Better hit ratio than direct mapping.
- Less expensive than fully associative mapping.

**Disadvantages:**

- More complex than direct mapping.
- Cost increases as the number of blocks per set increases.